



Build a Virtual Private Cloud



ajv.singh3107@gmail.com

VPC settings

Resources to create [Info](#)

Create only the VPC resource or the VPC and other networking resources.

☒ VPC only

☐ VPC and more

Name tag - *optional*

Creates a tag with a key of 'Name' and a value that you specify.

NextWork VPC

IPv4 CIDR block [Info](#)

☒ IPv4 CIDR manual input

☐ IPAM-allocated IPv4 CIDR block

IPv4 CIDR

10.0.0.0/16

CIDR block size must be between /16 and /28.

IPv6 CIDR block [Info](#)

☒ No IPv6 CIDR block

☐ IPAM-allocated IPv6 CIDR block

☐ Amazon-provided IPv6 CIDR block

☐ IPv6 CIDR owned by me

Tenancy [Info](#)

Default



Introducing Today's Project!

What is Amazon VPC?

Amazon VPC is a virtual network in AWS that allows you to isolate and control your cloud resources. It's useful because it provides flexibility to define IP ranges, subnets, security, and routing, ensuring secure, scalable, and customizable network.

How I used Amazon VPC in this project

I used Amazon VPC in today's project to create isolated subnets for public and private resources. I configured routing for secure internet access via an Internet Gateway, applied custom security groups, and ensured proper traffic flow between instanc

One thing I didn't expect in this project was...

One thing I didn't expect in this project was the complexity of configuring security groups. Balancing access permissions while ensuring security required careful planning, which highlighted the importance of network security in cloud environments.

This project took me...

It took me 40 minutes to complete this project.



Virtual Private Clouds (VPCs)

VPCs are Virtual Private Clouds in AWS, providing isolated network environments where users can launch and manage AWS resources securely. Each VPC can define IP ranges, subnets, routing, and gateways, offering control over traffic flow and network.

There was already a default VPC in my account ever since my AWS account was created. This is because AWS automatically provides a default VPC for each region to simplify the process of launching resources, allowing basic networking features.

To set up my VPC, I had to define an IPv4 CIDR, which means specifying a range of IP addresses using CIDR notation (e.g., 192.168.0.0/16). This defines the network size and allows efficient allocation of IP addresses within my VPC for resources.

The screenshot displays the 'VPC settings' page in the AWS Management Console. It includes sections for 'Resources to create' with radio buttons for 'VPC only' (selected) and 'VPC and more'; 'Name tag - optional' with a text input field containing 'NextWork VPC'; 'IPv4 CIDR block' with radio buttons for 'IPv4 CIDR manual input' (selected) and 'IPAM-allocated IPv4 CIDR block', followed by a text input field for the CIDR '10.0.0.0/16'; and 'IPv6 CIDR block' with radio buttons for 'No IPv6 CIDR block' (selected), 'IPAM-allocated IPv6 CIDR block', 'Amazon-provided IPv6 CIDR block', and 'IPv6 CIDR owned by me'. A note at the bottom states 'CIDR block size must be between /16 and /28'.

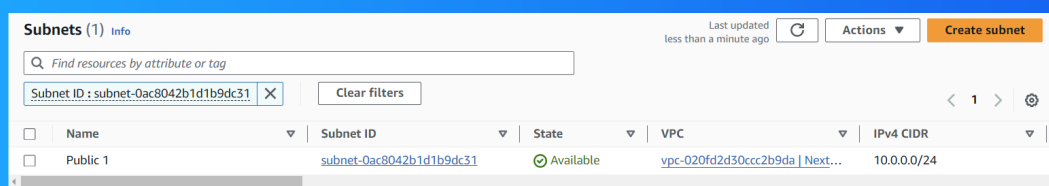


Subnets

Subnets are subdivisions of a VPC's IP address range that isolate resources within specific AZ. Each subnet allows control over traffic routing, security, and access, enabling more efficient network segmentation for public or private resource deploy.

There are already subnets existing in my account, one for every AZ in each region. AWS provides these default subnets to allow users to easily deploy resources across different zones with basic networking setup for high availability and redundancy.

I named my subnet Public 1, but that doesn't automatically make my subnet a public subnet. For a subnet to be considered public, it has to be associated with a route table that directs traffic through an Internet Gateway, allowing external access.



The screenshot shows the AWS Subnets console. At the top, it says 'Subnets (1)' with an 'info' link. Below this is a search bar and a 'Clear filters' button. A table lists the subnets. The first and only subnet is 'Public 1' with Subnet ID 'subnet-0ac8042b1d1b9dc31', State 'Available', VPC 'vpc-020fd2d30ccc2b9da', and IPv4 CIDR '10.0.0.0/24'.

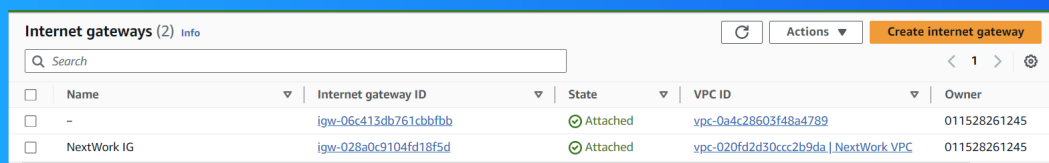
Name	Subnet ID	State	VPC	IPv4 CIDR
Public 1	subnet-0ac8042b1d1b9dc31	Available	vpc-020fd2d30ccc2b9da Next...	10.0.0.0/24



Internet gateways

Internet gateways are AWS components that allow communication between resources in a VPC and the internet. They provide a path for inbound and outbound traffic, enabling public access to instances within subnets configured with public IP addresses.

Attaching an internet gateway to a VPC means enabling the VPC to communicate with the internet. This allows resources, such as instances in public subnets, to send and receive traffic from external networks, making them accessible from outside VPC.



The screenshot shows the AWS Internet Gateways console. At the top, there's a header with 'Internet gateways (2)' and an 'Info' link. Below this is a search bar and a table of gateways. The table has columns for Name, Internet gateway ID, State, VPC ID, and Owner. Two gateways are listed: one with a hyphen as a name and another named 'NextWork IG'. Both are in an 'Attached' state. The 'NextWork IG' is attached to a VPC named 'NextWork VPC'.

Name	Internet gateway ID	State	VPC ID	Owner
-	igw-06c413db761cbbfbb	Attached	vpc-0a4c28603f48a4789	011528261245
NextWork IG	igw-028a0c9104fd18f5d	Attached	vpc-020fd2d30ccc2b9da NextWork VPC	011528261245



NextWork.org

Everyone should be in a job they love.

Check out nextwork.org for
more projects

